

Multi-level Attention-enhanced Learning for Fine-Grained Visual Classification

Kaifeng Ding

School of Computer Science and Technology(School of Artificial Intelligence), Zhejiang Normal University
Jinhua, ZheJiang, China
010711358@zjnu.edu.cn

Chengzhuan Yang*

School of Computer Science and Technology(School of Artificial Intelligence), Zhejiang Normal University
Jinhua, ZheJiang, China
czyang@zjnu.edu.cn

Caolin Yang

School of Computer Science and Technology(School of Artificial Intelligence), Zhejiang Normal University
Jinhua, ZheJiang, China
caolinyang@zjnu.edu.cn

Zhonglong Zheng

School of Computer Science and Technology(School of Artificial Intelligence), Zhejiang Normal University
Jinhua, ZheJiang, China
zhonglong@zjnu.edu.cn

Abstract

Due to significant intra-class variations and subtle inter-class differences, fine-grained images often pose challenges. In this paper, we propose a multi-level attention-enhanced network framework to address this issue. First, we employ convolutional features to crop the original images to focus on the main part and mitigate the interference of backgrounds on classification. Second, we design a scale-adaptive local discriminative region generation module based on the main part's images, effectively selecting locally detailed and rich image patches. It is worth noting that our proposed method only utilizes image-level labels. Finally, we extensively tested our method on three benchmark datasets, including the CUB-200-2011, Stanford Cars, and FGVC Aircraft. The experimental results demonstrate that our method achieves comparable or better classification performance than previous state-of-the-art fine-grained visual classification methods.

CCS Concepts

• **Computing methodologies** → **Object recognition; Machine learning.**

Keywords

Fine-grained Classification, Target Cropping, Discriminative Regions, Scale-adaptive, Weakly Supervised Learning

ACM Reference Format:

Kaifeng Ding, Caolin Yang, Chengzhuan Yang, and Zhonglong Zheng. 2024. Multi-level Attention-enhanced Learning for Fine-Grained Visual Classification. In *2024 8th International Conference on Computer Science and Artificial*

Intelligence (CSAI) (CSAI 2024), December 06–08, 2024, Beijing, China. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3709026.3709034>

1 Introduction

Fine-grained visual classification, distinct from traditional visual classification, involves the subdivision of categories within significant categories, such as birds[33], cars[19], and airplanes[27]. Fine-grained visual classification poses more significant challenges than traditional visual classification due to the significant intra-class variances and subtle inter-class differences exhibited by objects within the same category. Consequently, models trained on datasets like ImageNet[28], such as VGG[29], ResNet[10], and DenseNet[11], cannot be directly applied to fine-grained visual classification tasks. They require retraining on fine-grained visual classification datasets to achieve satisfactory performance, a process commonly referred to as transfer learning. Additionally, the limited size of some fine-grained visual datasets cannot support comprehensive deep learning training and necessitate transfer learning.

During the training process of fine-grained visual classification models, significant intra-class variations and minor inter-class differences pose a formidable challenge. As illustrated in Figures 1(a) and (b), although images belong to different categories, each column exhibits minimal differences. In contrast, those depicted in Figure 1(c) display substantial disparities despite sharing the same category label. In order to solve the above problems, previous research has proposed various methodologies, which we categorize into two main approaches: feature encoding-based methods[4, 7, 17, 20, 21, 41] and localization-based recognition methods[5, 6, 14, 18, 40, 43]. Feature encoding-based methods, exemplified by[21], leverage high-order pooling to facilitate learning semantically rich features, thereby enhancing classification accuracy. Moreover, the methods in [7, 17] have addressed the issue of excessively high-dimensional feature representations introduced by[21]. Additionally, Yu et al. [41] has identified the interplay between inter-layer feature interactions and fine-grained feature learning, indicating mutual reinforcement. In contrast, localization-based recognition methods, such as [1, 2, 14, 24, 37, 43], rely on manually annotated bounding box information, which is labor-intensive to acquire and hinders the scalability and application of fine-grained methods. Conversely, other studies [5, 6, 8, 18, 40, 42] have adopted a weakly

*Corresponding author

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CSAI 2024, December 06–08, 2024, Beijing, China

© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1818-2/24/12

<https://doi.org/10.1145/3709026.3709034>

supervised approach to explore the discriminative regions of images. Although employing weakly supervised learning mitigates the model’s dependence on manually labeled information, the precise localization of discriminative regions remains worthy of investigation.

This paper uses weakly supervised learning to effectively localize global and discriminative local regions within target images, utilizing solely image-level labels. We initially employed the pre-trained network model DenseNet[11] to extract feature maps from images. Subsequently, utilizing these feature maps, we determine the boundary coordinates of the target entities. Finally, building upon this initial target image processing, we employ a scale-adaptive local discriminative region generation module designed by us to identify the most discriminative local regions. In summary, the main contributions of our work are summarized as follows:

- Regarding global target localization, our approach exhibits superior versatility compared to [36] and [42], demonstrating robust localization performance.
- We introduce a scale-adaptive local discriminative region generation method that allows the model to automatically assess the saliency within the current region and determine whether to expand the search scope of local discriminative regions based on range thresholds, thereby achieving scale adaptability.
- Experimental results indicate that employing cross-entropy as the basis along with contrastive loss enhances the classification accuracy of fine-grained images.
- Our proposed method achieves performance on par with state-of-the-art approaches across three standard benchmark datasets.

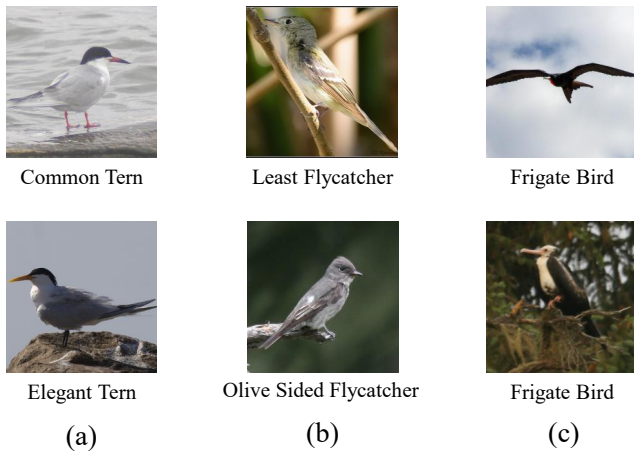


Figure 1: Fine-grained bird instances on the CUB-200-2011. The two columns, (a) and (b), represent birds with similar appearances but different categories, while column (c) illustrates the significant differences in the same bird species under different poses.

2 Related works

2.1 Feature Encoding-based Methods

Lin et al. [21] first introduced the concept of bilinear pooling in fine-grained visual classification. Given the outstanding performance exhibited by bilinear models in fine-grained classification, researchers have explored potential limitations from various perspectives. For instance, the methods in [7, 17, 41] respectively improve the standard bilinear model[21] from the perspectives of excessive feature dimensionality, computational demands, and inter-layer feature interactions. Song et al. [30] introduced global covariance pooling to fine-grained datasets by truncating small eigenvalues. However, they found that the model failed to converge after this operation. Experiment visualizations indicated that small eigenvalues were crucial for extracting discriminative class-specific features. Thus, inspired by these results, they proposed a network branch dedicated to preserving the importance of small eigenvalues. The commonality among these works lies in their encoding of convolutional features into high-order information. In contrast, Zhu et al. [47] proposed a new texture branch to supplement the CNN branch and address issues such as frequency bias and loss of detailed local information in CNNs. This texture branch enhances the method’s capability to effectively capture multi-frequency features and detailed local information. Liu et al. [23] proposed a novel invariant cues-aware feature concentration Transformer (TransIFC) for learning invariant and essential information in bird images. Xu et al. [38] proposed a novel internal ensemble learning transformer (IELT), which combines the concept of ensemble learning with Vision Transformer (ViT) to enhance fine-grained feature representation.

2.2 Localization-based Recognition Methods

We can divide localization-based recognition methods into supervised and weakly supervised methods. Supervised methods, such as those mentioned in [1, 2, 14, 24, 37, 43], rely on manually annotated localization boxes, which are costly and difficult to apply effectively. Therefore, current research generally relies on weakly supervised methods, which only use image-level labels to locate salient regions in images. For example, Fu et al. [6] argue that discriminative region localization and fine-grained feature learning are interrelated and can mutually reinforce each other. Therefore, they proposed a novel recursive attention convolutional neural network, which recursively learns discriminative region attention and region-based feature representation in a mutually reinforcing manner across multiple scales. Similarly, Yang et al. [40] designed a new training paradigm to enable navigation networks to detect regions with the maximum information under the guidance of teacher networks. Korsch et al. [18] proposed a novel local estimation method that uses initial predictions and gradient computations to determine feature importance and evaluates relevant image regions based on this. Recently, Ji et al. [15] designed a novel parallel multi-grained assembly of dual Transformer, exploring the information feature within various self-attention heads in Transformer layers and token assembly across Transformer layers. Tang et al. [32] proposed a novel fine-grained framework, PMRC(posture mining and reverse cross-entropy), to enhance the connections between different recognition regions within an entire image. This framework also mines

rich pose information within different discriminative regions to improve the model’s classification performance.

3 Method

In this section, we will provide a detailed description of the multi-level attention-enhanced learning framework. The overall framework overview is illustrated in Figure 2. We can mainly divide it into two modules: (1) The target cropping module utilizes the feature map extracted from the original images as the input of the method and can obtain the coordinates of the target object in the image as the output. (2) The scale-adaptive local discriminative region generation module considers the distribution of compressed feature values. It avoids the problem of selecting target images with high similarity by using peak suppression methods. Furthermore, considering that the details of different images vary in scale, making it impossible to use a fixed scale for capturing them, the method is designed in a scale-adaptive manner.

3.1 Target Cropping

The initial version of this method originated from [36] and was subsequently refined by [42], resulting in improved localization accuracy. In this paper, we streamline the approach proposed by [36, 42], utilizing only the last convolutional layer of DenseNet[11]. Although this modification may entail a slight sacrifice in localization accuracy, it enhances the method’s adaptability, making it easily applicable to other neural network models. The methodology begins by leveraging pre-trained neural network models to extract image features and obtain feature maps. Subsequently, feature aggregation is performed along the channel dimension, and a mask map is obtained by setting thresholds to delineate the boundary coordinates of the image’s main part. Finally, cropping is conducted to obtain the new image based on the derived coordinates. Specifically, we denote $F \in R^{C \times H \times W}$ as the output of the last convolutional layer of the convolutional neural network, representing the feature map of image X where C denotes the number of channels in the feature map and H and W represent the height and width of the feature map, respectively.

$$A = \sum_{i=0}^{C-1} f_i \quad (1)$$

Next, Equation (1) represents the specific operation of the feature map. Where f_i denotes the feature map of the i -th channel, thus, A is the result of aggregating the feature map F along the channel dimension.

$$\bar{a} = \frac{\sum_{x=0}^{W-1} \sum_{y=0}^{H-1} A(x, y)}{H \times W} \quad (2)$$

Subsequently, to obtain the mask map, we need to set a threshold represented by Equation (2). Equation (2) is derived from Equation (1) by averaging A over spatial scales to obtain the threshold $\bar{a}$. $\bar{a}$ determines the position of the main part, where (x, y) denotes a specific position on the size $H \times W$ feature map.

$$M = \begin{cases} 1 & \text{if } A > \bar{a} \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

Finally, we can utilize Equation (3) to obtain the required mask map from the last convolutional layer of DenseNet. By computing the maximum connected component of the mask map, we can

obtain the main part’s boundary coordinates, determine the main part’s position, and crop the target image accordingly. In [42], to enhance the localization accuracy, it utilizes the last convolutional layer of the last two convolutional blocks of ResNet[10] to obtain the feature maps conv5_b and conv5_c , respectively. Subsequently, equations (1), (2), and (3) were sequentially applied to derive the feature map $\tilde{M}_{\text{conv_5b}}$ and $\tilde{M}_{\text{conv_5c}}$, followed by equation (4) to derive the final mask map.

$$M = \tilde{M}_{\text{conv_5b}} \cap \tilde{M}_{\text{conv_5c}} \quad (4)$$

However, through experimentation, it was observed that once the method in the framework of [42] was modified, there was often little overlap between $\tilde{M}_{\text{conv_5b}}$ and $\tilde{M}_{\text{conv_5c}}$ during the training process, making it challenging to facilitate target localization better. Therefore, we simplified $\tilde{M}_{\text{conv_5b}}$ from [42] to address this issue and utilized only $\tilde{M}_{\text{conv_5c}}$ as the final mask map. Experimental results have demonstrated the effectiveness of this approach in localizing the target main part, achieving accuracy comparable to or slightly higher than [36], [44], and [3]. The specific performance is detailed in the next section.

3.2 Scale-adaptive Local Discriminative Region Generation

The method is proposed to address two issues: (1) How to obtain local discriminative regions beneficial for model discrimination. (2) How to deal with the varying scales of discriminative regions. Algorithm 1 details the execution flow of the core algorithm in this module.

The algorithm delineates a range by setting a range threshold, distinct from approaches employing fixed-scale exploration of discriminative regions. Since the maximum value of the feature map is not constant, this range is not fixed either. It will be adjusted based on training conditions. Our algorithm gradually approaches the threshold-defined range by expanding the boundary box range, hence called scale-adaptive. Inspired by peak suppression methods in [26, 31], we mask the selected region at the end of the algorithm. Peak suppression in [26, 31] aims to compel the network to explore alternative relevant regions by suppressing the most prominent areas. What is different is that we leverage the idea of peak suppression to restrain other sub-maximum values within the range of the maximum value threshold, mitigating excessive overlap in the discriminative regions we aim to locate, thereby providing the network with diverse, detailed images as much as possible.

3.3 Network Architecture and Loss Functions

The network designed in this paper consists of three branches: the original, global, and local branches. Firstly, the original branch processes the fine-grained images to obtain a feature map. Then, these feature maps are fed into the global branch through the target cropping module to extract the global features of the images. Finally, the scale-adaptive local discriminative region generation module is used to locate the discriminative regions of the images, and the local features of the images are obtained through the local branch. The loss functions for the three branches consist of cross-entropy loss and contrastive loss[9], as shown in Equations (5), (6), (7), and (8).

$$L_{\text{original}} = -\log(P_a(c)) \quad (5)$$

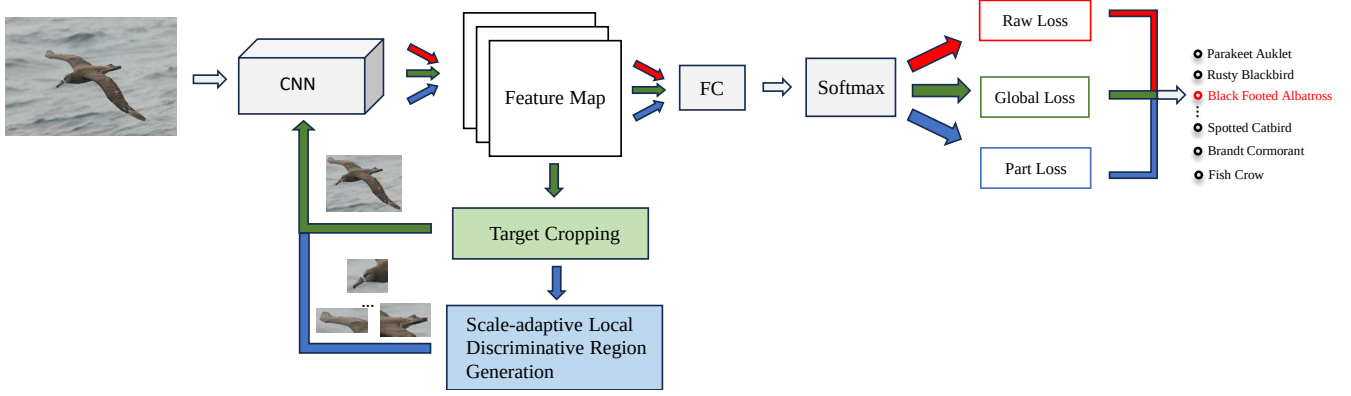


Figure 2: The overview of our overall architecture. The red branch is an original branch, the green branch is a global branch, and the blue branch is a local branch. Our method comprises two modules: (1) The Target Cropping module obtains the target image from the feature map of the original image; (2) The Scale-adaptive Local Discriminative Region Generation module utilizes the global features obtained from the Target Cropping module to generate multiple highly discriminative local images. The loss function of the network consists of both cross-entropy and contrastive loss.

Algorithm 1 Core Algorithm of Scale-adaptive Local Discriminative Region Generation Module

- 1: **Input:** The feature map of scale $C \times H \times W$, the range threshold, the number of discriminative regions (n_part)
- 2: **Output:** Coordinates of n_part bounding boxes (**Coords**)
- 3: Initialize bounding box coordinates: **Coords** $\leftarrow []$
- 4: By aggregating the feature map using Equation (1), a new feature map (**fm_sum**) with compressed channel dimensions is obtained, and the current scale of **fm_sum** is $H \times W$
- 5: **for** $i=0$ to n_part-1 **do**
- 6: Calculate the maximum value (**max_score**) and its corresponding indices of **fm_sum**
- 7: **current_score** $\leftarrow$ **max_score** $\times$ threshold
- 8: **new_score** $\leftarrow$ **current_score**
- 9: The variables **left_x**, **left_y**, **right_x**, and **right_y** represent the coordinates of the top-left and bottom-right corners of the bounding box, respectively, and (**left_x**, **left_y**) and (**right_x**, **right_y**) are initialized to the indices of the maximum values
- 10: **while** **current_score** $\leq$ **new_score** and ((**left_x** $>$ 0) or (**left_y** $>$ 0) or (**right_x** $<$ H) or (**right_y** $<$ W)) **do**
- 11: Expand the bounding box by one pixel
- 12: **new_score** $\leftarrow$ the mean value within the bounding box range
- 13: **end while**
- 14: **Coords** $\leftarrow$ [**left_x**, **left_y**, **right_x**, and **right_y**]
- 15: Mask the selected regions
- 16: **end for**
- 17: **return** **Coords**

$$L_{global} = -\log(P_b(c)) \quad (6)$$

$$L_{local} = -\sum_{n=0}^{N-1} \log(P_{p(n)}(c)) \quad (7)$$

$$L_{con} = \frac{1}{B^2} \sum_i^B \left[\sum_{j: y_i=y_j}^B (1 - \text{sim}(z_i, z_j)) + \sum_{j: y_i \neq y_j}^B \max(\text{sim}(z_i, z_j), 0) \right] \quad (8)$$

Where c represents the ground truth label of the input image, P_a and P_b represent the class probabilities output by the original branch and the global branch classifiers, respectively. $P_{p(n)}$ represents the class probability output by the n th image of the local branch classifier, where N is the number of discriminative regions. In the contrastive loss, B represents the number of image samples, and z_i and z_j are class-related feature maps that have been l_2 -normalized. Thus, $\text{sim}(z_i, z_j)$ is the dot product of z_i and z_j . The total loss of the model is formulated as shown in Equation (9).

$$L_{total} = L_{original} + L_{con_original} + L_{global} + L_{con_global} + L_{local} + L_{con_local} \quad (9)$$

Where $L_{con_original}$, L_{con_global} , and L_{con_local} represent the contrastive losses of the original, global, and local branches, respectively.

4 Experiments

4.1 Datasets

We conducted comprehensive experiments on three benchmark datasets: CUB-200-2011[33], FGVC Aircraft[27], and Stanford Cars[19], using only image-level labels. The detailed information of the three benchmark datasets is presented in Table 1.

Table 1: Details of the Three Benchmark Datasets

Datasets	Train	Test	Class
CUB-200-2011	5994	5794	200
FGVC Aircraft	6667	3333	100
Stanford Cars	8144	8041	196

4.2 Implementation Details

We preprocess the input images for the original branch and the global branch to the size of 448×448 , while resizing the input images for the local branch to 224×224 . The number of discriminative regions in the local branch is set to 7, and the range threshold is set to 0.8. We employ Momentum SGD with an initial learning rate of 0.001 and momentum of 0.9. The learning rate is decayed by a factor of 0.1 at 60 and 100 epochs, with weight decay set to $1e-4$. Training is conducted using a single NVIDIA GeForce RTX 3090 GPU, with a batch size of 6, and the model is trained for 200 epochs. Our method is implemented end-to-end, requiring no additional manual intervention.

4.3 Target Localization Performance

When evaluating the performance of the target cropping module, we employed the Percentage of Correctly Localized Parts (PCP) metric. It is defined as the percentage of predicted bounding boxes with an Intersection over Union (IOU) greater than 50% with the ground truth bounding boxes. As shown in Table 2, our streamlined approach still outperforms ACOL[44] and ADL[3] and is on par with the SCDA[36] method. Although it needs to catch up to the performance achieved by the AOLM[42] method, our approach exhibits higher applicability.

Table 2: Localization Performance of the Target Cropping Module on the CUB-200-2011 Dataset

Method	Loc Acc.(%)
ACOL[44]	46.0
ADL[3]	62.3
SCDA[36]	76.8
AOLM[42]	85.1
Ours	76.8

4.4 Quantitative Analysis

We compared our approach with previous works on three commonly used benchmark datasets, and the experimental results are presented in Table 3. While our method does not outperform all others on every dataset, it achieves comparable performance. Similarly, other methods exhibit a similar pattern, showing strengths on some datasets but relatively weaker on others.

From the third column of Table 3, it can be observed that the PMRC[32] method achieves the best performance on CUB-200-2011, reaching an impressive accuracy of 91.8%. While this represents a remarkable level of precision, it still requires training an additional deep navigation network to obtain discriminative regions in the images, thereby increasing the parameters of the network. In contrast, our scale-adaptive local discriminative region generation module does not increase the model’s parameters. The fourth and fifth columns of Table 3 present the performance of different methods on the FGVC Aircraft and Stanford Cars datasets, where our method competes favorably with state-of-the-art approaches or slightly outperforms them.

4.5 Ablation Experiments

We conduct ablation experiments on the CUB-200-2011, FGVC Aircraft, and Stanford Cars datasets to compare the impact of different model components on the classification results.

Target Cropping. Based on the performance shown in the second and third rows of Table 4, incorporating the target cropping module into the baseline DenseNet-161[11] demonstrates improved performance across three commonly used fine-grained visual classification datasets. Notably, it enhances the classification accuracy of the baseline model by 3% on the CUB[33] dataset, proving the effectiveness of this method. Additionally, the performance across the three datasets suggests that global features are relatively crucial for the CUB[33] dataset.

Scale-adaptive Local Discriminative Region Generation. Comparing the results in the second and fourth rows of Table 4, we observe that the local discriminative region generation module also universally enhances the model’s accuracy across the three datasets. This improvement is particularly evident on the CUB[33] dataset, where accuracy increases by 3%, demonstrating the effectiveness of this method.

Contrastive Loss. From the last two rows of Table 4, it is evident that the inclusion of contrastive loss improves the classification accuracy across all three datasets. The accuracy on the CUB[33], Air[27], and Car[19] datasets increases from 90.2% to 90.3%, 94.8% to 94.9%, and 95.4% to 95.5%, respectively. Experiments conducted on multiple datasets provide stronger evidence for the effectiveness of contrastive loss in enhancing the model’s performance.

Summary. First, from an overall perspective of Table 4, the baseline DenseNet-161 is more suitable for rigid objects. Second, from rows 3, 4, and 6, it can be observed that the accuracy improvements brought by the two modules are similar, and when combined, they can further enhance classification accuracy. Finally, it can be seen that the data for the Car dataset in rows 4 and 5 are the same. These observations indicate two points: (1) The local discriminative region generation module performs significantly well on the Car dataset. (2) It suggests that convolutional neural network-based models have less bias in classification accuracy when the number of training samples is large.

4.6 Qualitative Analysis

In this paper, we use red, blue, green, and yellow boxes to annotate the image regions of interest targeted by the scale-adaptive local discriminative region generation module. As shown in Figure 3, the module focuses on the beak and feathers for birds, the head and windows for cars, and the tail wing and turbines for airplanes. These focal points resemble human attention patterns.

5 Conclusion

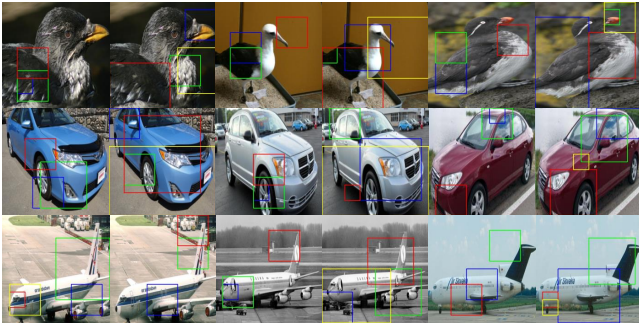
In this paper, we propose a multi-level attention-enhanced architecture. It consists of two components: (1) Target Cropping Module, which captures the main parts of the image through the feature map of the original image, obtaining a global feature map; (2) Scale-adaptive local discriminative region generation module, which automatically captures highly discriminative local images at different scales, generating multiple local feature map. Neither

Table 3: Performance Comparison of CNN-Based Approaches on CUB-200-2011, FGVC Aircraft, and Stanford Cars

Method	Backbone	CUB Acc.(%)	Air Acc.(%)	Car Acc.(%)
ACNet[16]	ResNet-50	88.1	92.4	94.6
PMG[5]	ResNet-50	89.6	93.4	95.1
DF-GMM[35]	ResNet-50	88.8	93.8	94.8
FDL[22]	DenseNet-161	89.1	91.3	94.0
API-Net[48]	DenseNet-161	90.0	93.9	95.3
MMAL[42]	ResNet-50	89.6	94.7	95.0
GHORD[46]	ResNet-50	89.6	94.3	95.1
DP-Net[34]	ResNet-50	89.3	93.9	94.8
SPS[13]	ResNet-50	88.7	92.7	94.9
PART[45]	ResNet-50	89.6	94.4	95.1
CLAN[12]	ResNet-50	88.1	93.2	94.2
DTRG[25]	DenseNet-161	90.1	94.8	95.7
P2P-Net[39]	ResNet-50	90.2	94.2	95.4
PMRC[32]	ResNet-50	91.8	94.8	95.4
Ours	ResNet-50	89.1	94.6	95.2
Ours	DenseNet-161	<u>90.3</u>	94.9	<u>95.5</u>

Table 4: Ablation Study to Explore the Effects of Different Components of Our Method on the Classification Results of the CUB-200-2011, FGVC Aircraft, and Stanford Cars Datasets

Method	Target Cropping	Scale-adaptive Local Discriminative Region Generation	Contrastive Loss	CUB Acc.(%)	Air Acc.(%)	Car Acc.(%)
Ours			✓	85.5	93.1	94.5
	✓		✓	88.5	94.1	95.1
		✓	✓	88.5	93.9	95.4
	✓	✓		90.2	94.8	95.4
	✓	✓	✓	90.3	94.9	95.5

**Figure 3: The image regions attended by the proposed scale-adaptive local discriminative region generation module.**

of the two modules increases the model’s parameters and only employs image-level labels. Experimental results on three benchmark datasets demonstrate that it achieves comparable or better classification performance than the prior state-of-the-art methods. Future work will continue to explore improving the localization accuracy of both global and local images. This would enable the network to learn more precise key features sufficient to distinguish between different categories, thereby enhancing classification accuracy.

Acknowledgments

This work was supported by the Key Projects of Jinhua Science and Technology Bureau (Project no. 2024-2-015), the National Natural Science Foundation of China (Grant No. 62106227), the China Postdoctoral Science Foundation (Grant No. 2023M743132), and the "Teacher Professional Development Project" for Domestic Visiting Scholars in 2023 (Project No. FX2023007).

References

- [1] Steve Branson, Grant Van Horn, Serge Belongie, and Pietro Perona. 2014. Bird species categorization using pose normalized deep convolutional nets. *arXiv preprint arXiv:1406.2952* (2014).
- [2] Yuning Chai, Victor Lempitsky, and Andrew Zisserman. 2013. Symbiotic segmentation and part localization for fine-grained categorization. In *Proceedings of the IEEE International Conference on Computer Vision*. 321–328.
- [3] Junsuk Choe and Hyunjung Shim. 2019. Attention-based dropout layer for weakly supervised object localization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2219–2228.
- [4] Yin Cui, Yang Song, Chen Sun, Andrew Howard, and Serge Belongie. 2018. Large scale fine-grained categorization and domain-specific transfer learning. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 4109–4118.
- [5] Ruoyi Du, Dongliang Chang, Ayan Kumar Bhunia, Jiyang Xie, Zhanyu Ma, Yi-Zhe Song, and Jun Guo. 2020. Fine-grained visual classification via progressive multi-granularity training of jigsaw patches. In *European Conference on Computer Vision*. Springer, 153–168.
- [6] Jianlong Fu, Heliang Zheng, and Tao Mei. 2017. Look closer to see better: Recurrent attention convolutional neural network for fine-grained image recognition.

- In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 4438–4446.
- [7] Yang Gao, Oscar Beijbom, Ning Zhang, and Trevor Darrell. 2016. Compact bilinear pooling. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 317–326.
 - [8] Weifeng Ge, Xiangru Lin, and Yizhou Yu. 2019. Weakly supervised complementary parts models for fine-grained image classification from the bottom up. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3034–3043.
 - [9] Ju He, Jie-Neng Chen, Shuai Liu, Adam Kortylewski, Cheng Yang, Yutong Bai, and Changhu Wang. 2022. Transfg: A transformer architecture for fine-grained recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 852–860.
 - [10] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 770–778.
 - [11] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger. 2017. Densely connected convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 4700–4708.
 - [12] Ranran Huang, Yu Wang, and Huazhong Yang. 2022. Cross-layer attention network for fine-grained visual categorization. *arXiv preprint arXiv:2210.08784* (2022).
 - [13] Shaoli Huang, Xinchao Wang, and Dacheng Tao. 2021. Stochastic partial swap: Enhanced model generalization and interpretability for fine-grained recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 620–629.
 - [14] Shaoli Huang, Zhe Xu, Dacheng Tao, and Ya Zhang. 2016. Part-stacked CNN for fine-grained visual categorization. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 1173–1182.
 - [15] Ruyi Ji, Jiaying Li, Libo Zhang, Jing Liu, and Yanjun Wu. 2023. Dual transformer with multi-grained assembly for fine-grained visual classification. *IEEE Transactions on Circuits and Systems for Video Technology* (2023).
 - [16] Ruyi Ji, Longyin Wen, Libo Zhang, Dawei Du, Yanjun Wu, Chen Zhao, Xianglong Liu, and Feiyue Huang. 2020. Attention convolutional binary neural tree for fine-grained visual categorization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 10468–10477.
 - [17] Shu Kong and Charless Fowlkes. 2017. Low-rank bilinear pooling for fine-grained classification. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 365–374.
 - [18] Dimitri Korsch, Paul Bodesheim, and Joachim Denzler. 2019. Classification-specific parts for improving fine-grained visual categorization. In *Pattern Recognition: 41st DAGM German Conference, DAGM GCPR 2019, Dortmund, Germany, September 10–13, 2019, Proceedings 41*. Springer, 62–75.
 - [19] Jonathan Krause, Michael Stark, Jia Deng, and Li Fei-Fei. 2013. 3d object representations for fine-grained categorization. In *Proceedings of the IEEE international conference on computer vision workshops*. 554–561.
 - [20] Zhichao Li, Yi Yang, Xiao Liu, Feng Zhou, Shilei Wen, and Wei Xu. 2017. Dynamic computational time for visual attention. In *Proceedings of the IEEE international conference on computer vision workshops*. 1199–1209.
 - [21] Tsung-Yu Lin, Aruni Roy Chowdhury, and Subhransu Maji. 2015. Bilinear CNN models for fine-grained visual recognition. In *Proceedings of the IEEE international conference on computer vision*. 1449–1457.
 - [22] Chuanbin Liu, Hongtao Xie, Zheng-Jun Zha, Lingfeng Ma, Lingyun Yu, and Yongdong Zhang. 2020. Filtration and distillation: Enhancing region attention for fine-grained visual categorization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 34. 11555–11562.
 - [23] Hai Liu, Cheng Zhang, Yongjian Deng, Bochen Xie, Tingting Liu, Zhaoli Zhang, and You-Fu Li. 2023. TransIFC: invariant cues-aware feature concentration learning for efficient fine-grained bird image classification. *IEEE Transactions on Multimedia* (2023).
 - [24] Jiongxin Liu, Angjoo Kanazawa, David Jacobs, and Peter Belhumeur. 2012. Dog breed classification using part localization. In *Computer Vision–ECCV 2012: 12th European Conference on Computer Vision, Florence, Italy, October 7–13, 2012, Proceedings, Part I 12*. Springer, 172–185.
 - [25] Kangjun Liu, Ke Chen, and Kui Jia. 2022. Convolutional fine-grained classification with self-supervised target relation regularization. *IEEE Transactions on Image Processing* 31 (2022), 5570–5584.
 - [26] Xinda Liu, Lili Wang, and Xiaoguang Han. 2022. Transformer with peak suppression and knowledge guidance for fine-grained image recognition. *Neurocomputing* 492 (2022), 137–149.
 - [27] Subhransu Maji, Esa Rahtu, Juho Kannala, Matthew Blaschko, and Andrea Vedaldi. 2013. Fine-grained visual classification of aircraft. *arXiv preprint arXiv:1306.5151* (2013).
 - [28] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, et al. 2015. Imagenet large scale visual recognition challenge. *International journal of computer vision* 115 (2015), 211–252.
 - [29] Karen Simonyan and Andrew Zisserman. 2014. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556* (2014).
 - [30] Yue Song, Nicu Sebe, and Wei Wang. 2022. On the eigenvalues of global covariance pooling for fine-grained visual recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 3 (2022), 3554–3566.
 - [31] Guolei Sun, Hisham Cholakkal, Salman Khan, Fahad Khan, and Ling Shao. 2020. Fine-grained recognition: Accounting for subtle differences between similar classes. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 34. 12047–12054.
 - [32] Zhenchao Tang, Hualin Yang, and Calvin Yu-Chian Chen. 2023. Weakly Supervised Posture Mining for Fine-Grained Classification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 23735–23744.
 - [33] C. Wah, S. Branson, P. Welinder, P. Perona, and S. Belongie. 2011. *The Caltech-UCSD birds-200-2011 dataset*. Technical Report CNS-TR-2011-001. California Institute of Technology.
 - [34] Shijie Wang, Haojie Li, Zhihui Wang, and Wanli Ouyang. 2021. Dynamic position-aware network for fine-grained image recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35. 2791–2799.
 - [35] Zhihui Wang, Shijie Wang, Shuhui Yang, Haojie Li, Jianjun Li, and Zezhou Li. 2020. Weakly supervised fine-grained image classification via gaussian mixture model oriented discriminative learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 9749–9758.
 - [36] Xiu-Shen Wei, Jian-Hao Luo, Jianxin Wu, and Zhi-Hua Zhou. 2017. Selective convolutional descriptor aggregation for fine-grained image retrieval. *IEEE transactions on image processing* 26, 6 (2017), 2868–2881.
 - [37] Lingxi Xie, Qi Tian, Richang Hong, Shuicheng Yan, and Bo Zhang. 2013. Hierarchical part matching for fine-grained visual categorization. In *Proceedings of the IEEE international conference on computer vision*. 1641–1648.
 - [38] Qin Xu, Jiahui Wang, Bo Jiang, and Bin Luo. 2023. Fine-grained visual classification via internal ensemble learning transformer. *IEEE Transactions on Multimedia* (2023).
 - [39] Xuhui Yang, Yaowei Wang, Ke Chen, Yong Xu, and Yonghong Tian. 2022. Fine-grained object classification via self-supervised pose alignment. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 7399–7408.
 - [40] Ze Yang, Tiange Luo, Dong Wang, Zhiqiang Hu, Jun Gao, and Liwei Wang. 2018. Learning to navigate for fine-grained classification. In *Proceedings of the European conference on computer vision (ECCV)*. 420–435.
 - [41] Chaojian Yu, Xinyi Zhao, Qi Zheng, Peng Zhang, and Xinge You. 2018. Hierarchical bilinear pooling for fine-grained visual recognition. In *Proceedings of the European conference on computer vision (ECCV)*. 574–589.
 - [42] Fan Zhang, Meng Li, Guisheng Zhai, and Yizhao Liu. 2021. Multi-branch and multi-scale attention learning for fine-grained visual categorization. In *MultiMedia Modeling: 27th International Conference, MMM 2021, Prague, Czech Republic, June 22–24, 2021, Proceedings, Part I 27*. Springer, 136–147.
 - [43] Ning Zhang, Jeff Donahue, Ross Girshick, and Trevor Darrell. 2014. Part-based R-CNNs for fine-grained category detection. In *Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part I 13*. Springer, 834–849.
 - [44] Xiaolin Zhang, Yunchao Wei, Jiashi Feng, Yi Yang, and Thomas S Huang. 2018. Adversarial complementary learning for weakly supervised object localization. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 1325–1334.
 - [45] Yifan Zhao, Jia Li, Xiaowu Chen, and Yonghong Tian. 2021. Part-Guided Relational Transformers for Fine-Grained Visual Recognition. *IEEE Transactions on Image Processing* 30 (2021), 9470–9481.
 - [46] Yifan Zhao, Ke Yan, Feiyue Huang, and Jia Li. 2021. Graph-based high-order relation discovery for fine-grained recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 15079–15088.
 - [47] Lanyun Zhu, Tianrun Chen, Jianxiong Yin, Simon See, and Jun Liu. 2023. Learning gabor texture features for fine-grained recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 1621–1631.
 - [48] Peiqin Zhuang, Yali Wang, and Yu Qiao. 2020. Learning attentive pairwise interaction for fine-grained classification. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 34. 13130–13137.